

EXP NO : 3

25/07/2026

IMPLEMENTING ASSET MANAGEMENT WITH HYPERLEDGER FABRIC

Aim:

To implement an Asset Management application using Hyperledger Fabric by creating, reading, updating, transferring, and deleting assets through JavaScript chaincode and verifying the transactions on the blockchain ledger.

Requirements

- Kali Linux / Ubuntu
- Docker and Docker Compose
- Node.js (Version 18 or later) and npm
- Git
- Hyperledger Fabric Samples, Binaries, and Docker Images

Procedure & Execution

1. Navigate to the Test

Network Command(s):

```
(student@linux)-[~]  
$ cd ~/fabric-dev/fabric-samples/test-network
```

2. Start the Hyperledger

Fabric Network Command(s):

```
(student@linux)-[~]  
$ ./network.sh up createChannel -ca  
  
-----  
Creating channel 'mychannel'  
Starting peer0.org1  
Starting peer0.org2  
Starting orderer.example.com  
Network Started Successfully
```

3. Deploy the Asset

Transfer Chaincode Command(s):

```
(student@linux)-[~]  
$ ./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript  
  
-----  
Packaging Chaincode  
Installing Chaincode  
Approving Chaincode  
Committing Chaincode  
Chaincode committed successfully
```

4. Initialize the Ledger

Command(s):

NAME: RENUGA DEVI K

ROLL NO: 231901041

```
(student@linux) - [ ]  
$
```

```
(student@linux) - [ ]
```

```
└─$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --
```

Ledger Initialized Successfully

5. Query All Assets

Command(s):

```
(student@linux)-[~]
$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'

[
  {
    "ID": "asset1",
    "Color": "blue",
    "Owner": "Tomoko",
    "Size": 5,
    "AppraisedValue": 300
  },
  {
    "ID": "asset2",
    "Color": "red",
    "Owner": "Brad",
    "Size": 5,
    "AppraisedValue": 400
  }
]
```

6. Create a New Asset

Command(s):

```
(student@linux) - [ ]
```

```
└─$ peer chaincode invoke -o localhost:7050 -- ordererTLSHostnameOverride orderer.example.com
```

```
Asset asset11 created successfully
```

7. Read the Asset Details

Command(s):

```
(student@linux)-[~]
$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset11"]}'
-----
{
  "ID": "asset11",
  "Color": "White",
  "Size": 12,
  "Owner": "User",
  "AppraisedValue": 900
}
```

8. Update Asset

Information Command(s):

```
(student@linux)-[~]
$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --
cafile $ORDERER_CA -C mychannel -n basic -c '{"function":"UpdateAsset", "Args":
["asset11","Black","15","User","1200"]}'
-----
Asset Updated Successfully
```

9. Transfer Asset

Ownership Command(s):

```
(student@linux)-[ ]
```

```
└─$ peer chaincode invoke -o localhost:7050 -- ordererTLSHostnameOverride orderer.example.com -- tls --  
-----
```

Ownership transferred successfully

10. Verify the Updated Asset

Command(s):

```
(student@linux)-[~]  
$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset11"]}'  
-----  
{  
  "ID": "asset11",  
  "Color": "Black",  
  "Size": 15,  
  "Owner": "David",  
  "AppraisedValue": 1200  
}
```

11. Delete the Asset

Command(s):

```
(student@linux) - [ ]
```



```
└─$ peer chaincode invoke -o localhost:7050 -- ordererTLSHostnameOverride orderer.example.com --  
-----
```

Asset Deleted Successfully

12. Verify Asset Deletion

Command(s):

```
(student@linux)-[~]  
$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset", "asset11"]}'  
-----  
Error: asset11 does not exist
```

13. Stop the Network

Command(s):

```
(student@linux)-[~]  
$ ./network.sh down  
-----  
Removing containers  
Removing Docker Network  
Network stopped successfully
```

Result:

The Asset Management application was successfully implemented using Hyperledger Fabric. The asset lifecycle operations, including asset creation, querying, updating, ownership transfer, and deletion, were executed successfully. The transactions were recorded on the blockchain ledger, demonstrating secure and immutable asset management in a permissioned blockchain network.